CORE PURE / POLAR COORDINATES

Polar Coordinates Basics

Questions by topic 10 questions · 97 marks

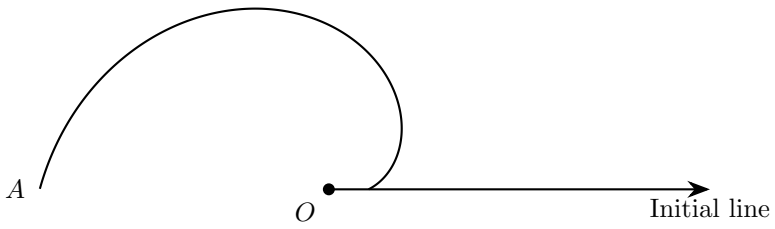
EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

This pack takes the polar radius to be non-negative, $r \geq 0$. OCR MEI (H645) also sketches curves with $r < 0$, drawn with a broken line, which this pack does not cover.

Questions

Question 1	*8 marks*	2
Question 2	*8 marks*	3
Question 3	*7 marks*	4
Question 4	*8 marks*	5
Question 5	*11 marks*	6
Question 6	*7 marks*	7
Question 7	*12 marks*	8
Question 8	*13 marks*	9
Question 9	*8 marks*	10
Question 10	*15 marks*	11



1. The polar curves C_1 and C_2 are defined by

$$C_1 : r = 1 + 2\theta, \qquad 0 \leq \theta \leq \pi$$

and

$$C_2 : r = 8 - \theta, \qquad 0 \leq \theta \leq 2\pi$$

The diagram above shows C_1. Its endpoint A lies on the ray opposite the initial line.

(a) State the polar coordinates of A and find its Cartesian coordinates, taking the pole as the origin and the initial line as the positive x-axis. **[3]**

(b) Find the polar coordinates of the point where C_1 and C_2 intersect. **[2]**

(c) On a copy of the diagram above, sketch C_2, indicating the radii of its two endpoints. **[3]**

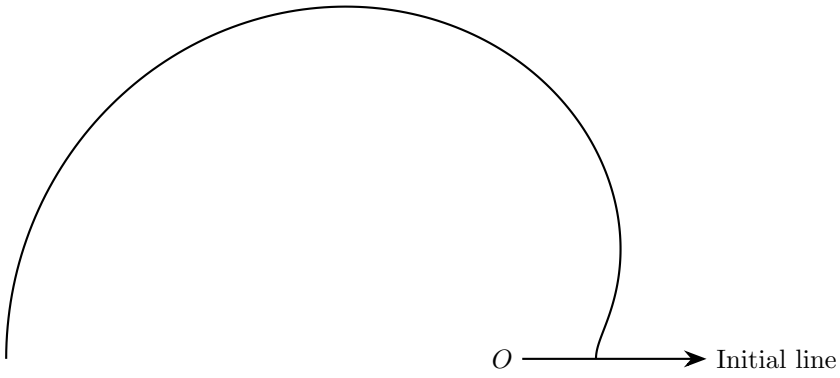
2. The curve C has the polar equation

$$r = \frac{6}{2 - \sin\theta}, \qquad -\pi < \theta \leq \pi$$

(a) Verify that the point with polar coordinates $\left(4, \frac{\pi}{6}\right)$ lies on C. **[1]**

(b) Find the exact polar coordinates of the point on C which is furthest from the pole, O. **[3]**

(c) The Cartesian line $y = x$ intersects C at two points. Find the exact Cartesian coordinates of both points. **[4]**



3. The curve C has polar equation

$$r = 4 - 3\cos\theta, \qquad -\pi < \theta \leq \pi$$

(a) The diagram above shows the portion of C for which $0 \leq \theta \leq \pi$. Complete the sketch of C. **[1]**

The straight line ℓ has polar equation

$$r\cos\theta = 1$$

(b) Add ℓ to your completed sketch and find all the points of intersection of C and ℓ. Give their polar coordinates, with radii exact and any non-zero angles in radians to 3 significant figures. **[6]**

4. The curve C_1 has polar equation

$$r^2 + 4r\cos\theta - 6r\sin\theta = 12$$

where $r > 0$ and $-\pi < \theta \leq \pi$.

(a) Find a Cartesian equation for C_1 and show that it represents a circle. State its radius and the Cartesian coordinates of its centre. **[4]**

(b) Sketch C_1, marking its centre and the pole. Find the exact greatest and least distances of a point on C_1 from the pole. **[4]**

5. **(a)** The curve C_1 has Cartesian equation

$$x^2 + y^2 = \sqrt{3}\, y$$

Find the centre and radius of C_1, and show that its non-pole points have polar equation $r = \sqrt{3}\sin\theta$. **[4]**

The curve C_2 has polar equation

$$r = 1 + \cos\theta$$

Consider both curves for $0 < \theta < \pi$, together with their limiting endpoints.
In this interval, $r > 0$ for both curves. The pole is a limiting endpoint of each curve.

(b) Show that C_1 and C_2 have exactly one intersection other than the pole, and find the exact polar coordinates of this intersection, P. **[4]**

(c) Sketch both curves on a single diagram, clearly labelling each curve. Mark P and indicate the Cartesian coordinates of every point where either curve meets the initial line. **[3]**

6. A curve has Cartesian equation

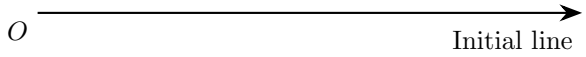
$$(x^2 + y^2)^2 = 5x^2 + 4xy + 2y^2$$

where x and y are not both zero. Use the convention $r > 0$.

(i) Show that the polar equation of the curve is

$$r^2 = \frac{7 + 3\cos 2\theta + 4\sin 2\theta}{2}$$ **[4]**

(ii) Hence find the least and greatest distances of points on the curve from the origin, giving your answers in exact form. **[3]**



7. The graph G_1 has polar equation

$$r = 6\sec\left(\theta + \frac{\pi}{6}\right), \qquad -\frac{2\pi}{3} < \theta < \frac{\pi}{3}$$

The graph G_2 is the ray with polar equation

$$\theta = -\alpha, \qquad r > 0$$

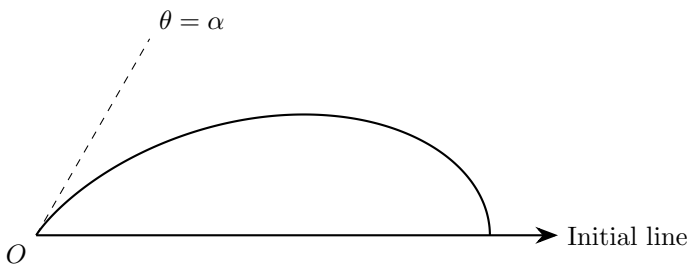
where $0 < \alpha < \frac{\pi}{2}$.

(a) Show that G_1 is a straight line by finding its Cartesian equation in the form $y = mx + c$. **[3]**

(b) On a copy of the diagram above, sketch G_1 and G_2. Label the angle α and give the polar coordinates of the intersection of G_1 with the initial line. **[3]**

(c) Find, in terms of α, the polar coordinates of the intersection of G_1 and G_2. **[2]**

(d) The region bounded by the initial line, G_1 and G_2 has area $12\sqrt{3}$ square units. Determine the exact value of α. **[4]**



8. A curve has polar equation

$$r = 2(1 + \cos 3\theta), \qquad 0 \leq \theta < 2\pi$$

with only points for which $r > 0$ represented by polar coordinates. The pole O is included as a limiting point of the curve.

(i) The diagram above shows the arc traced from $\theta = 0$ until the curve first reaches O. The tangent there has equation $\theta = \alpha$, where $0 < \alpha < \pi$. Find α in terms of π. **[2]**

(ii) (a) Let $f(\theta) = 2(1 + \cos 3\theta)$. Show that, for every integer k,

$$f\left(\frac{2k\pi}{3} - \theta\right) = f(\theta)$$

and also show that

$$f\left(\theta + \frac{2\pi}{3}\right) = f(\theta)$$ **[4]**

(b) Hence state the equations of all the distinct lines of symmetry of the curve, using angles in $0 \leq \theta < \pi$. State also the smallest positive angle through which the curve can be rotated about O and remain unchanged. **[3]**

(iii) Sketch the complete curve. State its maximum radius and all the corresponding values of θ in $0 \leq \theta < 2\pi$. **[4]**

9. **(a)** The line L has polar equation

$$r = 3\sec\left(\theta - \frac{\pi}{4}\right), \qquad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

Show that its Cartesian equation is $x + y = 3\sqrt{2}$. **[2]**

(b) The curve C has polar equation

$$r = 5 - 2\sin 2\theta, \qquad -\pi < \theta \leq \pi$$

Find the exact polar coordinates of all the points of intersection of L and C.
Fully justify that you have found every point of intersection. **[6]**

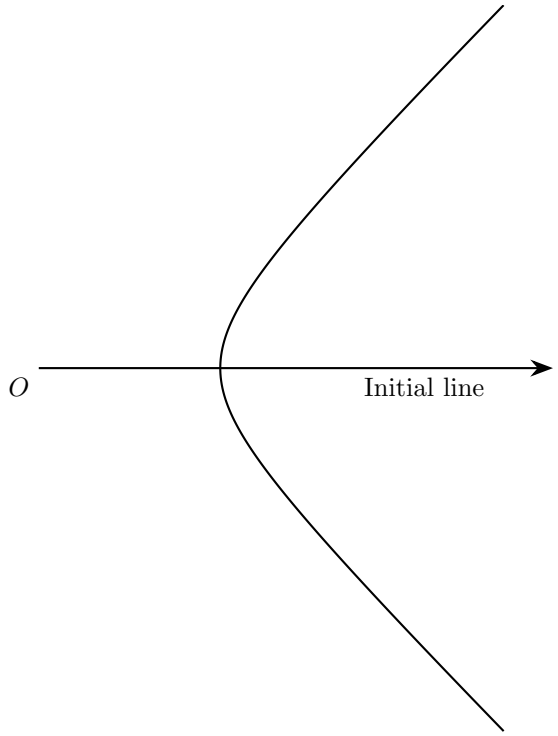
10. **(a)** The curve C is the right-hand branch of the hyperbola

$$x^2 - y^2 = 6x, \qquad x \geq 6$$

Using $x = r\cos\theta$ and $y = r\sin\theta$, show that its polar equation is

$$r = 6\cos\theta\sec 2\theta, \qquad -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

where $r > 0$. **[3]**



(b) The diagram above shows the curve C.

(i) Find the polar coordinates of the point N on C nearest to the pole O. Justify that your point gives the minimum value of r. **[3]**

(ii) The circle $r = 6\sqrt{3}$ meets C at P and Q, where P lies above the initial line and Q lies below it. Find their exact polar coordinates, using angles in the interval $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. **[5]**

(iii) Find the exact value, in radians, of the angle PNQ. **[4]**